

DAY 05 — HTTP Request Inspection

Assessment Information

Organization: Trios Cyber

Assignment: HTTP Request Inspection

Day: 05

Application: Damn Vulnerable Web Application (DVWA)

Target: 192.168.56.101

Tool: Burp Suite

Environment: Authorized local VirtualBox laboratory

1. Executive Summary

Day 05 focused on understanding HTTP request and response traffic using Burp Suite.

Burp Suite was configured as a proxy between the browser and the authorized DVWA laboratory application. HTTP traffic generated during normal application navigation and login activity was captured and inspected.

A total of 10 HTTP requests were documented. The captured traffic was analyzed for HTTP methods, request paths, parameters, headers, cookies, and response status codes.

No exploitation, brute-force activity, payload testing, or destructive actions were performed.

2. Assessment Objectives

The objectives of this assessment were to:

1. Configure Burp Suite as an HTTP proxy.
 2. Configure the browser to send traffic through Burp Suite.
 3. Capture at least 10 HTTP requests from DVWA.
 4. Identify HTTP request methods.
 5. Identify request paths and parameters.
 6. Inspect HTTP headers.
 7. Identify cookies used by the application.
 8. Identify HTTP response status codes.
 9. Document the captured traffic professionally.
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3. Authorization and Scope

The activity was performed exclusively against the authorized local DVWA laboratory system.

Target: 192.168.56.101

In Scope

- Burp Suite proxy configuration
- Browser proxy configuration
- Normal DVWA navigation
- HTTP request inspection
- HTTP response inspection
- Header and cookie analysis

Out of Scope

- Exploitation
 - Brute-force attacks
 - SQL injection testing
 - Command injection
 - File inclusion exploitation
 - Credential attacks
 - Vulnerability exploitation
 - Persistence
 - Privilege escalation
 - Denial-of-service activity
 - Destructive actions
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4. Lab Environment

Workstation

- Operating System: Kali Linux
- Network Interface: eth0
- Kali IP: 192.168.56.102/24

Target

- Application: Damn Vulnerable Web Application (DVWA)
- Target IP: 192.168.56.101
- Network: VirtualBox Host-Only laboratory network

Tool

- Burp Suite
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5. Burp Suite Proxy Configuration

Burp Suite was configured to operate as the interception proxy for browser traffic.

Evidence: Burp Suite proxy setup

The browser was configured to route HTTP traffic through the Burp Suite proxy.

Evidence: Browser proxy configuration

6. HTTP Request Capture

Ten HTTP requests were captured from the DVWA application during normal navigation.

Request 1

Method: GET

Path: /

Response: 200 OK

Evidence: HTTP request 1

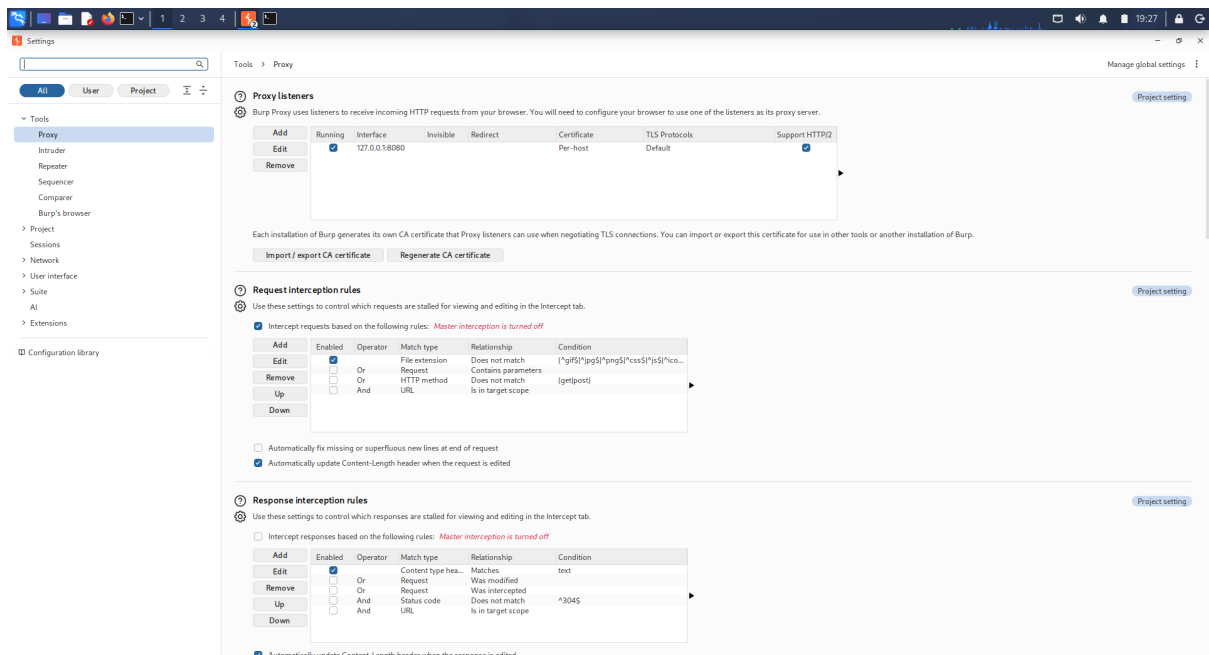


Figure 1: Burp Suite proxy setup

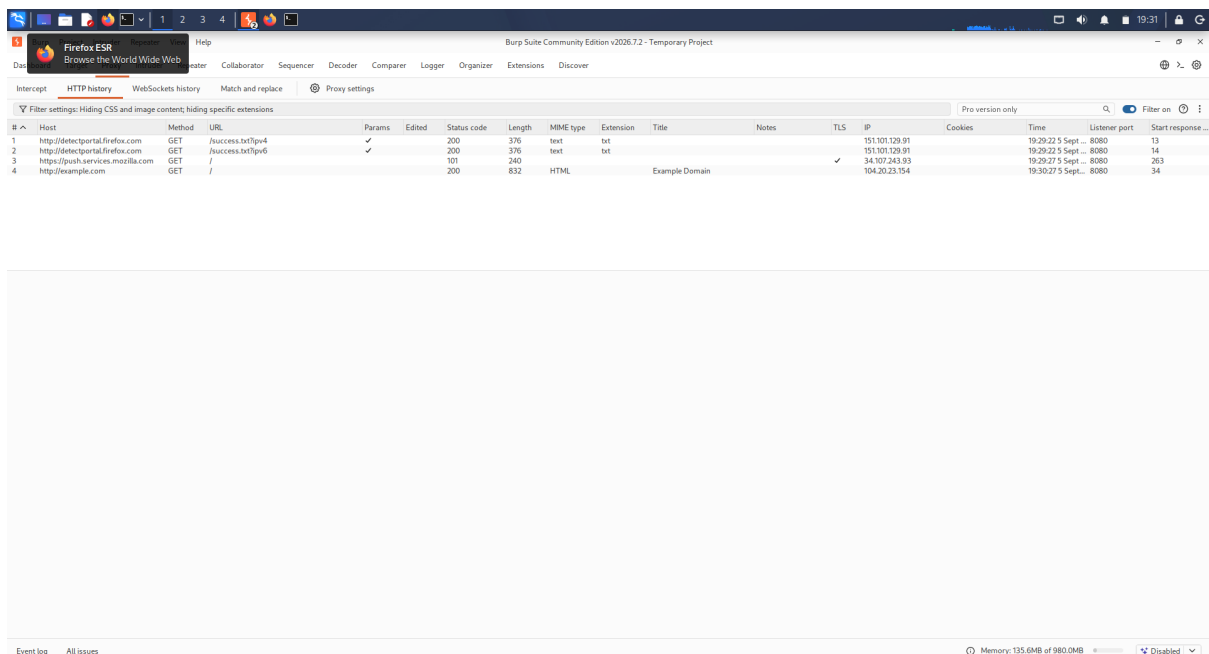


Figure 2: Browser proxy configuration

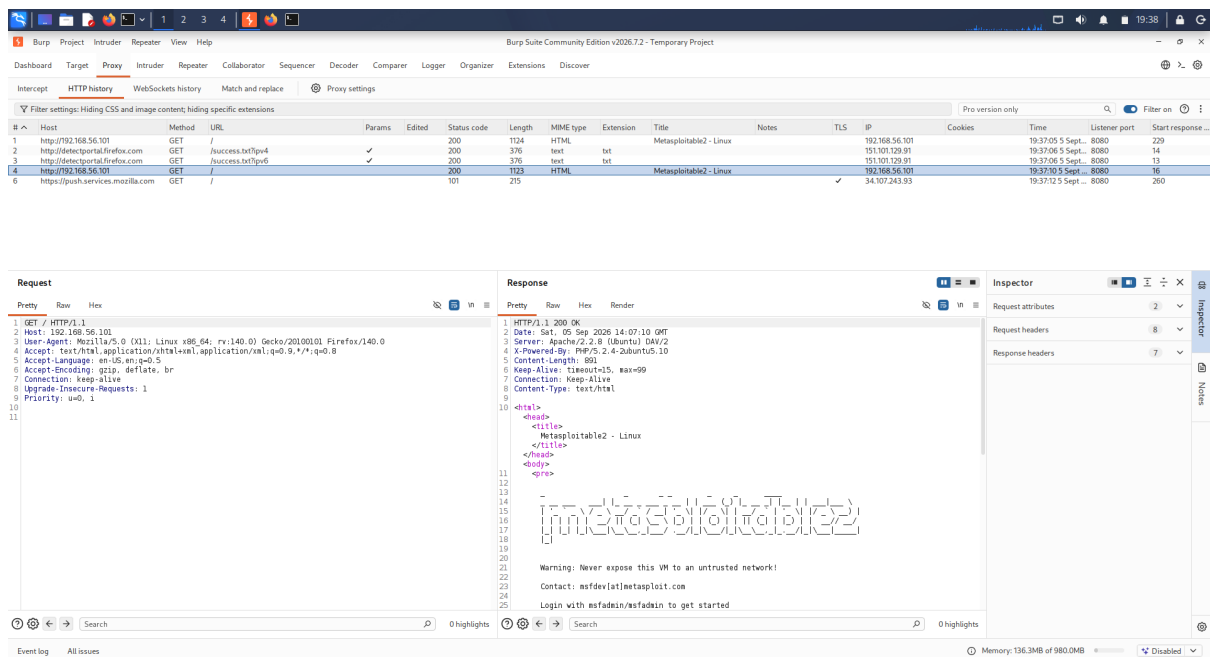


Figure 3: HTTP request 1

Request 2

Method: GET

Path: /

Response: 200 OK

Evidence: HTTP request 2

Request 3

Method: GET

Path: /dvwa/login.php

Response: 200 OK

Evidence: DVWA login page request

Request 4

Method: POST

Path: /dvwa/login.php

Response: 302 Found

The login request demonstrated the use of a POST method and form parameters.

Evidence: Login POST request

Request 5

Method: GET

Path: /dvwa/index.php

Response: 200 OK

Evidence: DVWA index request

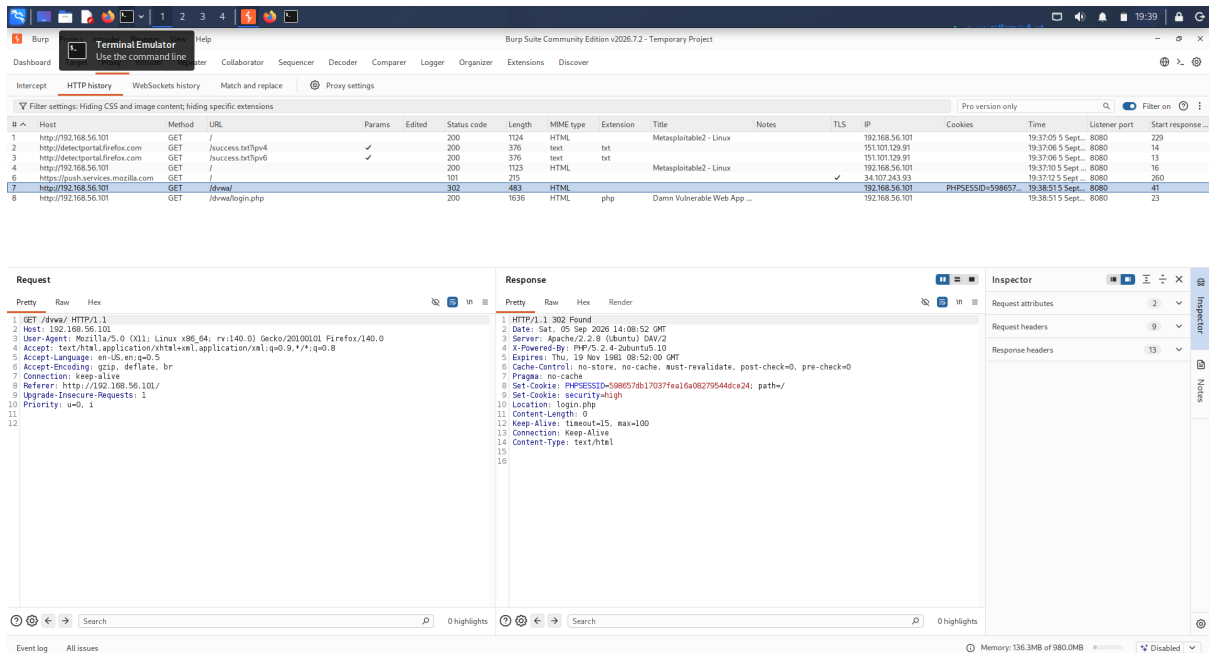


Figure 4: HTTP request 2

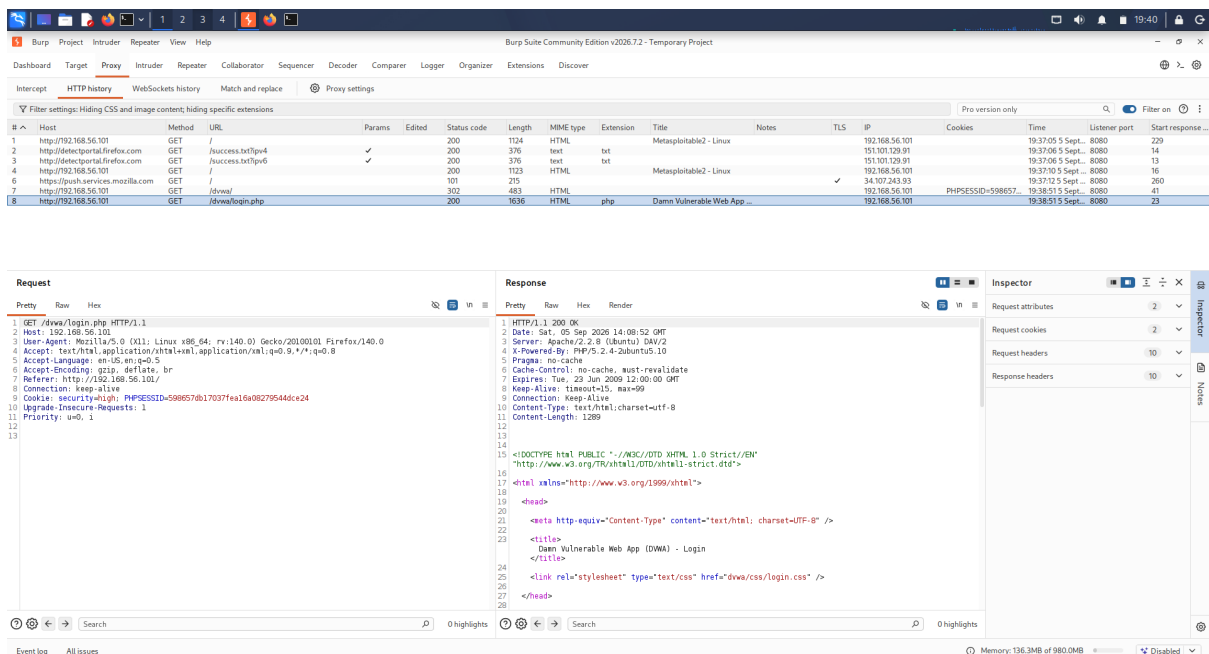


Figure 5: DVWA login page request

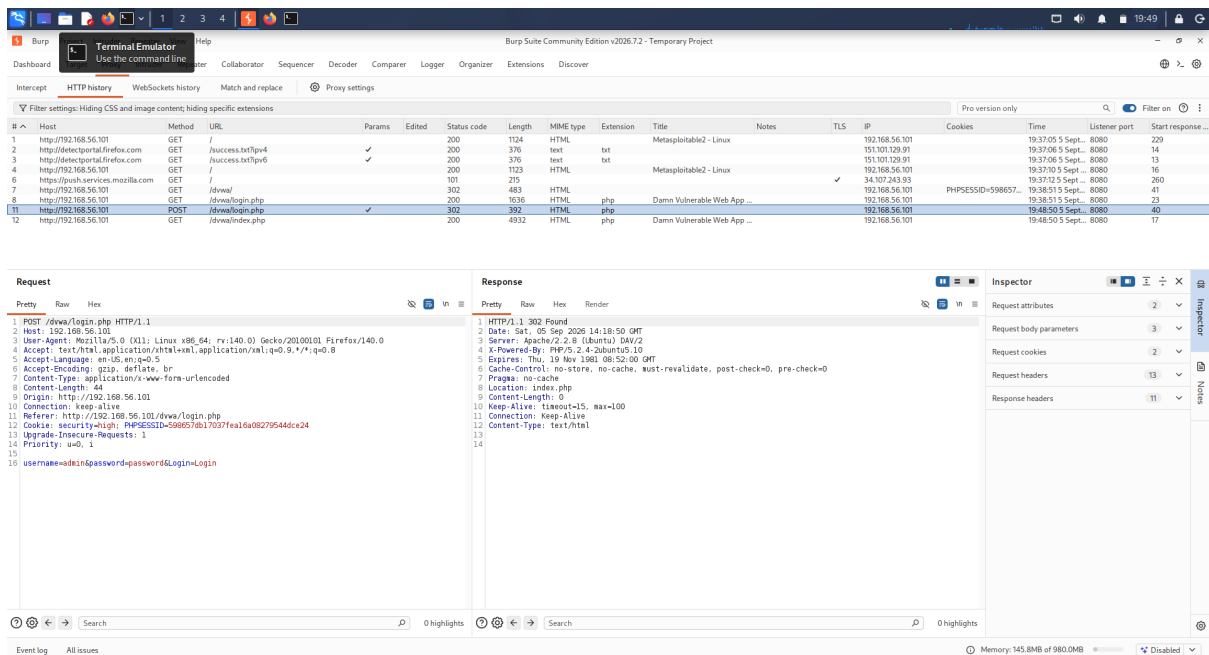


Figure 6: Login POST request

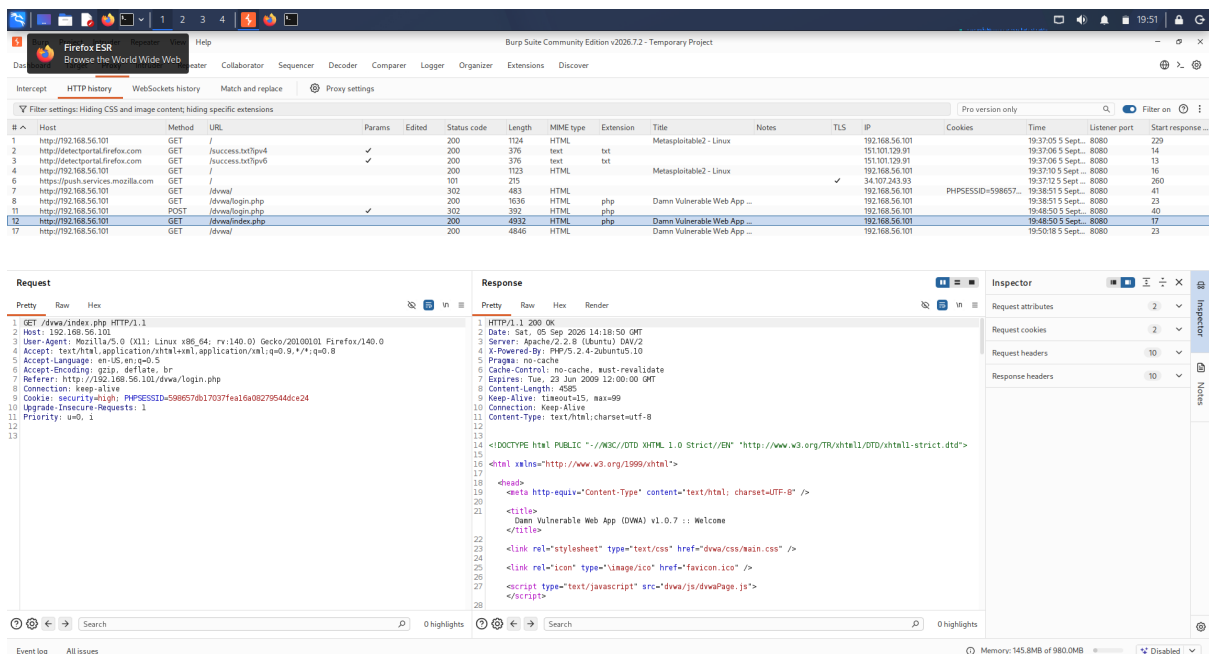


Figure 7: DVWA index request

Request 6

Method: GET

Path: /dvwa/instructions.php

Response: 200 OK

Evidence: Instructions request

The screenshot shows the Burp Suite interface. The top panel displays the HTTP history table with 18 entries. The selected entry (index 18) is a GET request to /dvwa/instructions.php. The bottom panel shows the details of this request and response.

#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Cookies	Time	Listener port	Start response...
1	http://192.168.56.101	GET	/			200	1124	HTML		Metasploitable2 - Linux			192.168.56.101		19.37.05 5 Sept.	8080	229
2	http://detectportal.firefox.com	GET	/success.txt?pv=4			200	376	text	txt				151.101.129.91		19.37.06 5 Sept.	8080	14
3	http://detectportal.firefox.com	GET	/success.txt?pv=6			200	376	text	txt				151.101.129.91		19.37.06 5 Sept.	8080	13
4	http://192.168.56.101	GET	/			200	1123	HTML		Metasploitable2 - Linux			192.168.56.101		19.37.10 5 Sept.	8080	16
6	http://push.services.mozilla.com	GET	/			101	275						34.107.243.93		19.37.12 5 Sept.	8080	260
7	http://192.168.56.101	GET	/dvwa/			302	483	HTML					192.168.56.101	PHPSESSID=598657...	19.38.51 5 Sept.	8080	41
8	http://192.168.56.101	GET	/dvwa/login.php			200	1636	php		Damn Vulnerable Web App ...			192.168.56.101		19.38.51 5 Sept.	8080	23
11	http://192.168.56.101	POST	/dvwa/login.php			302	392	HTML					192.168.56.101		19.48.50 5 Sept.	8080	40
12	http://192.168.56.101	GET	/dvwa/index.php			200	4932	HTML	php	Damn Vulnerable Web App ...			192.168.56.101		19.48.50 5 Sept.	8080	17
17	http://192.168.56.101	GET	/dvwa/			200	4846	HTML		Damn Vulnerable Web App ...			192.168.56.101		19.50.18 5 Sept.	8080	23
18	http://192.168.56.101	GET	/dvwa/instructions.php			200	10108	HTML	php	Damn Vulnerable Web App ...			192.168.56.101		19.51.48 5 Sept.	8080	32

The details panel shows the request and response for the selected entry. The request is a GET request to /dvwa/instructions.php. The response is a 200 OK status with an HTML body containing the instructions for the DVWA application.

Figure 8: Instructions request

Request 7

Method: GET

Path: /dvwa/vulnerabilities/brute/

Response: 200 OK

The page was accessed only for normal request inspection. No brute-force activity was performed.

Evidence: Brute Force page request

Request 8

Method: GET

Path: /dvwa/vulnerabilities/exec/

Response: 200 OK

The page was accessed only for normal HTTP inspection. No command injection payloads were submitted.

Evidence: Command Injection page request

Request 9

Method: GET

Path: /dvwa/vulnerabilities/fi/

Response: 200 OK

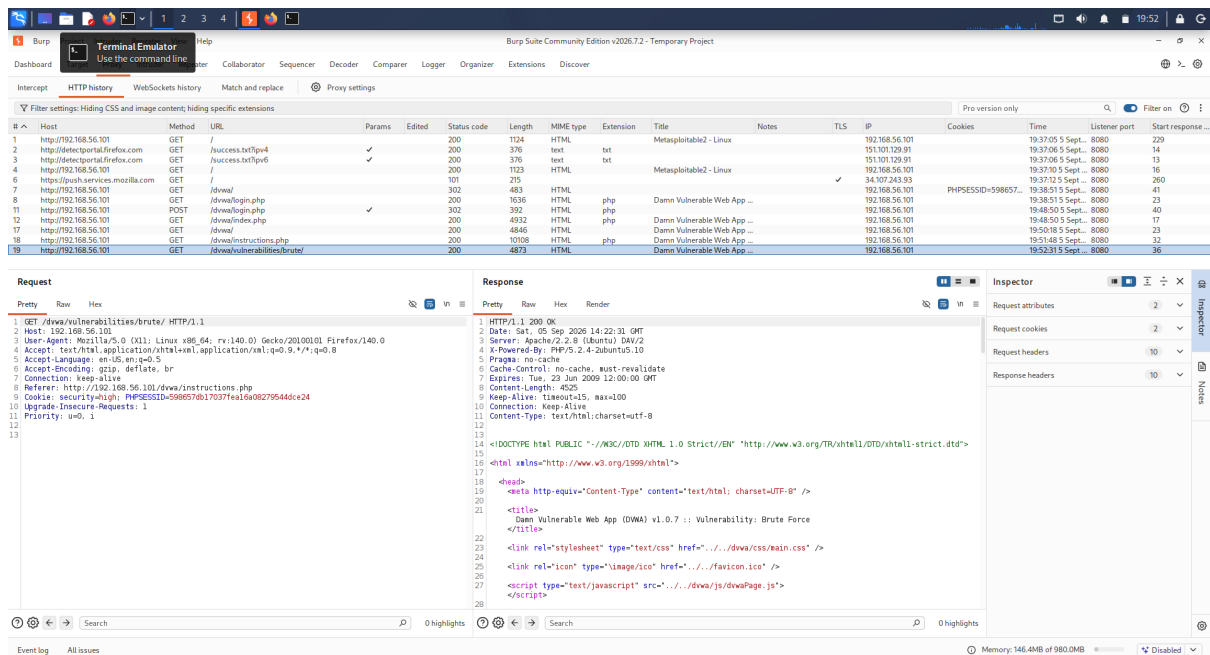


Figure 9: Brute Force page request

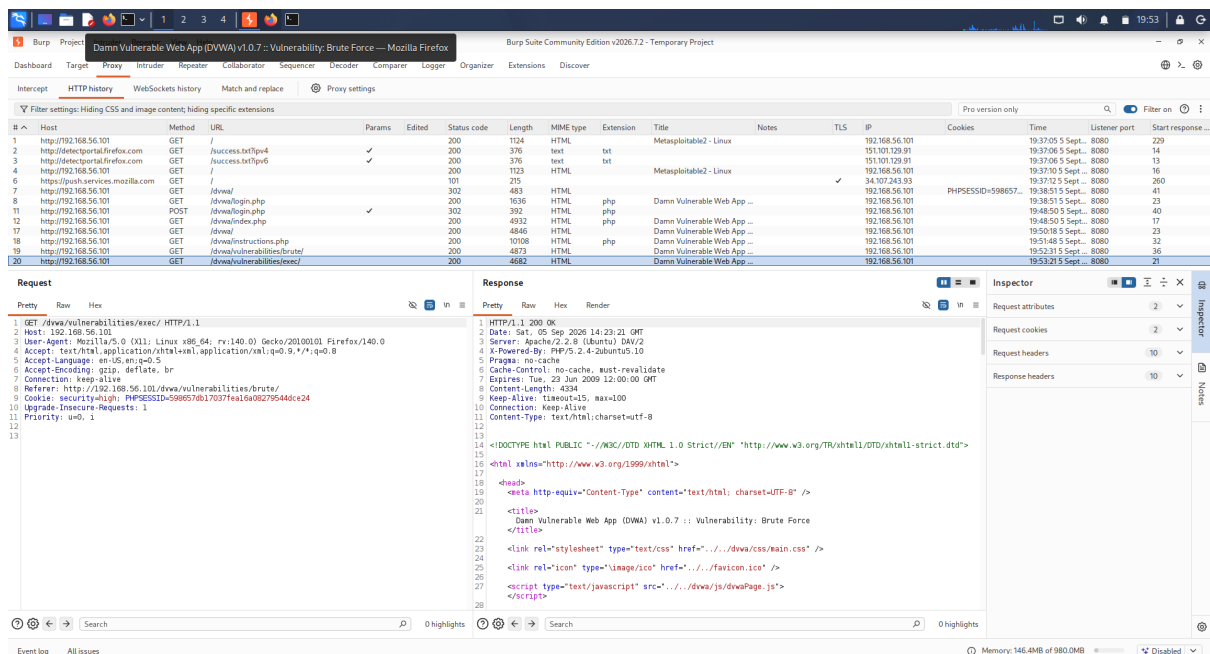


Figure 10: Command Injection page request

The page was accessed only for normal HTTP inspection. No file inclusion payloads were submitted.

Evidence: File Inclusion page request

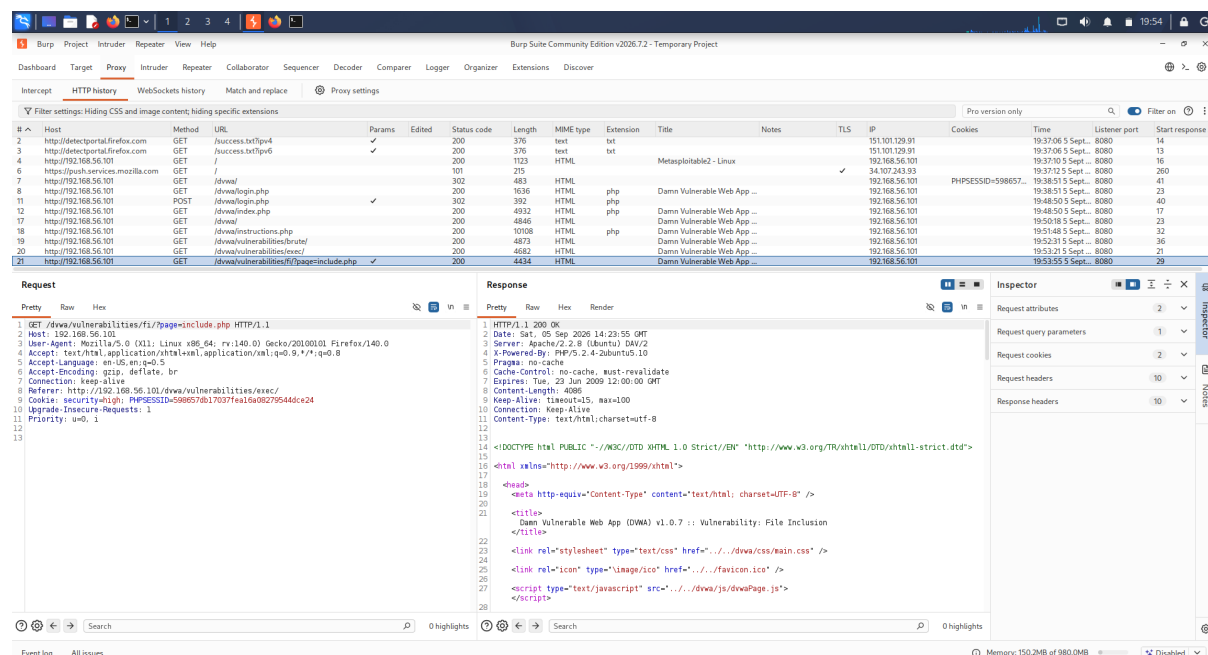


Figure 11: File Inclusion page request

Request 10

Method: GET

Path: /dvwa/vulnerabilities/sqli/

Response: 200 OK

The page was accessed only for normal HTTP inspection. No SQL injection payloads were submitted.

Evidence: SQL Injection page request

7. HTTP Methods Observed

The captured traffic demonstrated two HTTP methods.

GET

GET requests were used to retrieve web pages and application resources.

POST

A POST request was observed during the DVWA login process:

POST /dvwa/login.php

The POST request contained form parameters associated with the login process.

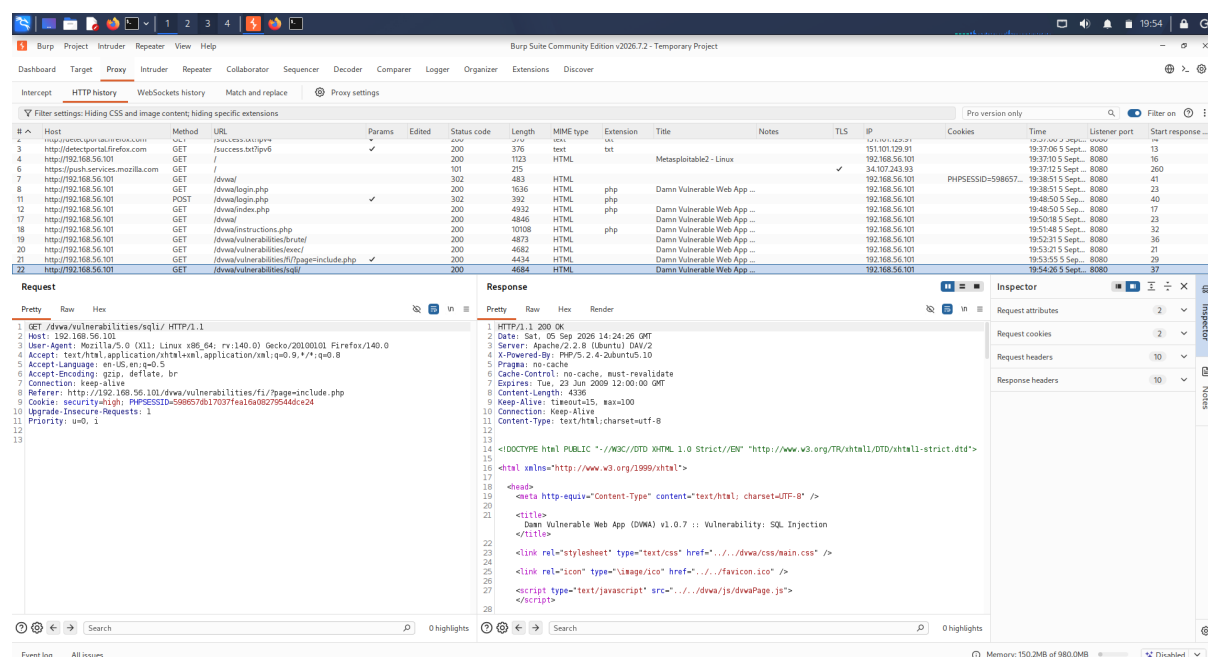


Figure 12: SQL Injection page request

8. Parameters

Burp Suite allowed request parameters to be inspected.

The login POST request contained authentication form parameters. Sensitive session values and credentials are not reproduced in this report.

9. HTTP Headers

Burp Suite provided visibility into HTTP request and response headers.

Examples include:

- Host
- User-Agent
- Accept
- Content-Type
- Referer
- Cookie
- Content-Length

Headers provide information about the client, requested resource, request format, and session state.

10. Cookies

The DVWA application used session-related cookies during authenticated navigation.

A session cookie was observed in the captured traffic.

Actual session cookie values are not reproduced in this report.

11. Response Status Codes

Status Code	Meaning	Observed Use
200	OK	Successful page requests
302	Found / Redirect	Login/navigation redirect

12. Request Summary

#	Method	Path	Response
1	GET	/	200
2	GET	/	200
3	GET	/dvwa/login.php	200
4	POST	/dvwa/login.php	302
5	GET	/dvwa/index.php	200
6	GET	/dvwa/instructions.php	200
7	GET	/dvwa/vulnerabilities/brute/	200
8	GET	/dvwa/vulnerabilities/exec/	200
9	GET	/dvwa/vulnerabilities/fi/	200
10	GET	/dvwa/vulnerabilities/sqli/	200

13. Evidence Index

Evidence	Description
screenshots/01-burp-proxy-setup.png	Burp Suite proxy setup
screenshots/02-browser-proxy.png	Browser proxy configuration
screenshots/03-request-01.png	HTTP request 1
screenshots/04-request-02.png	HTTP request 2
screenshots/05-request-03-login.png	DVWA login page request
screenshots/06-request-04-login-post.png	Login POST request
screenshots/07-request-05-index.png	DVWA index request
screenshots/08-request-06-instructions.png	Instructions request
screenshots/09-request-07-brute-force-page.png	Brute Force page request
screenshots/10-request-08-command-injection.png	Command Injection page request
screenshots/11-request-09-file-inclusion.png	File Inclusion page request
screenshots/12-request-10-sql-injection.png	SQL Injection page request

14. Learning Outcomes

- Configured Burp Suite as an HTTP proxy.
- Configured browser proxy settings.
- Captured browser traffic with Burp Suite.
- Identified GET and POST methods.
- Inspected request paths.
- Identified request parameters.
- Inspected HTTP request and response headers.
- Understood session cookies.

- Interpreted HTTP response status codes.
 - Practiced analyzing web application traffic in an authorized laboratory environment.
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15. Security and Ethical Considerations

All testing was performed against the authorized local DVWA laboratory environment.

The vulnerable application was used strictly for educational HTTP traffic inspection.

No unauthorized systems were targeted, and no exploitation or destructive testing was performed.

Session information and sensitive values were not reproduced in the final report.

16. Conclusion

Day 05 successfully demonstrated HTTP request inspection using Burp Suite.

Burp Suite was configured between the browser and DVWA, allowing HTTP traffic to be captured and analyzed. Ten requests were documented and examined for methods, paths, parameters, headers, cookies, and response status codes.

The exercise provided practical understanding of how web browsers communicate with web applications and how a security analyst can inspect that communication using a proxy.

Assessment Completion Status

Assessment Status: Completed

Requests Captured: 10

Evidence Status: Complete

Scope Status: Authorized Local Laboratory Only

Report Status: Ready for PDF Generation and Git Submission